

Faculty To-Do Lists to Help Students Upskill with AI (CSU East Bay)

Department of Statistics & Data Science — CSU East Bay

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1 Purpose

A concise, faculty-facing plan—derived from the Provost’s **Back to the Bay 2025** keynote—covering key ideas and concrete actions we can embed in Statistics and Data Science courses to **upskill** students.

Reference keynote: *Back to the Bay Keynote Script – AI + Workforce Development*, August 13–14, 2025. Office of Academic Affairs, CSU East Bay. (link in **References**).

2 Executive Summary of the Provost’s Vision

2.1 What the Provost Means by “Upskilling”

- **Deliberately adding new capabilities and mindsets** that prepare students to **apply knowledge in unfamiliar situations**—not just listing more skills on a résumé.
- Goal: Graduates who **learn quickly, communicate effectively, exercise judgment, and work across disciplines** in a world where **the only constant is change**.
- Motivation: AI is **redefining “entry-level”** work; employers expect more **complex, judgment-based** tasks earlier in careers.

2.2 Three-Part Framework

1. **AI Fluency** — Understand how AI works, what it can and **cannot** do, and how to apply it **responsibly**.
2. **Human-Centric Skills** — Empathy, ethics, communication, teamwork, leadership, and professionalism that **AI cannot replace**.
3. **Agility** — The capacity to **adapt quickly** when conditions change and to course-correct with confidence.

3 Main Areas the Keynote Highlighted for Higher Education

- **Demographic & trust headwinds** for higher education; we must **prove value** and improve student **readiness** earlier in the curriculum.
- **AI adoption in industry** is accelerating; “entry-level” expectations are moving toward **judgment, communication, and collaboration**.
- We must intentionally cultivate: **AI fluency, human-centric skills, and agility** across courses, co-curriculars, community projects, and research.

4 Main Areas for the Future Job Market (CSUEB Students)

- Employers seek more than technical competence: **critical thinking**, **clear communication to diverse audiences**, **ethical judgment**, **collaboration with people and digital tools**, **professionalism**, **curiosity**, and **relational intelligence**.
- **Experience gap** matters as much as skills: students need **realistic, messy contexts** (people, deadlines, shifting priorities) to practice judgment.

5 Faculty To-Do Lists by Area

5.1 1. AI Fluency — What It Means

For students: - Baseline literacy in **model capabilities/limits**, **data provenance**, **prompting & verification**, **privacy & security**, and **responsible use**. - Ability to **choose and justify** when to use (or not use) AI tools.

For faculty: - Integrate **scaffolded, low-stakes uses** of AI with explicit **guardrails**, **attribution**, and **evaluation** standards.

5.1.1 Faculty To-Do (Statistics & Data Science)

- **Syllabus & Policy**
 - Add an **AI use policy**: permitted tools (e.g., ChatGPT Edu, code copilots), **disclosure/attribution**, **verification steps**, and **privacy** expectations.
 - Create a **simple decision rubric** (“use AI / do not use AI / use with verification”) for common task types.
- **In-Class Micro-Labs (15–25 min)**
 - Prompt-engineering warmups: design and test prompts for **EDA**, **data cleaning**, **simulation setup**, then **compare outputs**.
 - **Model limits** demo: adversarial cases (e.g., extrapolation beyond data, spurious correlations) and **error analysis**.
- **Assignments**
 - Require **AI-aided** and **AI-free** versions of selected tasks; students submit a **tool log** and a **verification note**.
 - Add **reproducibility** checkpoints: students must **re-run code** end-to-end; attach a **session info** block.
- **Assessment**

- Include **viva-style checks** or short **code-explanations** to validate understanding beyond AI outputs.
- Use **rubrics** that reward **method choice**, **assumption checks**, and **interpretation**, not just final answers.

5.2 2. Human-Centric Skills — What It Means

For students: - Practice **empathy**, **ethics**, **teamwork**, **leadership**, and **clear communication** under constraints and stakeholder diversity.

For faculty: - Design tasks where **the human lens is essential**: ambiguity, trade-offs, equity, and audience-aware storytelling.

5.2.1 Faculty To-Do (Statistics & Data Science)

- **Communication Across Audiences**
 - Require a **two-pass deliverable**: (a) technical memo; (b) **one-page brief** for a **non-technical** stakeholder.
 - Add **visual narrative** checks: students must justify figure choices (markings, scales, uncertainty displays).
- **Ethics & Equity**
 - Short ethics prompts tied to content (e.g., **bias in training data**, disparate impact, data privacy). Students propose **mitigations**.
 - Include a **model risk statement** with each project: assumptions, failure modes, **potential harm**, and **monitoring**.
- **Teamwork & Professionalism**
 - Structured team roles: **Lead**, **Reviewer**, **Communicator**, **Scribe**—rotate mid-project.
 - Use **issue trackers**/Kanban for deadlines; grade **process artifacts** (meeting notes, task boards).
- **Mentoring Moments**
 - Implement **peer code-reviews** and **peer feedback** rounds with a short **reflective memo** on received feedback.

5.3 3. Agility — What It Means

For students: - Comfort with **changing requirements**, **new data**, and **iteration**; ability to **pivot** while preserving rigor.

For faculty: - Create **time-boxed sprints**, **mid-stream requirement changes**, and **A/B comparisons** of solution paths.

5.3.1 Faculty To-Do (Statistics & Data Science)

- **Agility Sprints**
 - Run 1–2 week sprints where the **data generating process** shifts mid-sprint.
 - Include a **retro**: what changed, how the team adapted, what to automate next time.
- **Model-Change Drills**
 - Provide a baseline model; then supply **new features** or **shifted distributions** and assess **update speed** and **validation**.
- **Tooling**
 - Encourage **notebooks + scripts + Makefiles** (or **targets** in R) for **re-runnable pipelines** under change.

6 Experiential Learning Activities (by Area)

Design for real-world context, stakeholders, deadlines, and evolving constraints.
Where AI assists, require **verification and reflection**.

6.1 AI Fluency — Activity Ideas

- **AI-Aided EDA & Verification Lab**

Students use AI to propose EDA steps on a noisy dataset; they must **verify** at least three AI claims with code and explain any discrepancies.
- **Baseline vs. AI-Assisted Forecasting**

Compare a classical model and an AI-assisted approach on a time series. Evaluate with $RMSE = \sqrt{\frac{1}{n} \sum_i (\hat{y}_i - y_i)^2}$ and report **pros/cons**.
- **Prompt-to-Pipeline**

Start from an English prompt; produce a minimal, reproducible R pipeline (e.g., **targets**) with **unit tests** for key functions.

6.2 Human-Centric Skills — Activity Ideas

- **Stakeholder Brief + Technical Appendix**

Teams analyze a dataset with equity implications (e.g., admissions, lending). Deliver: (1) a **one-page policy brief**; (2) a **technical appendix** with sensitivity analysis and uncertainty intervals $\left[\hat{\theta} \pm z_{0.975} \widehat{\text{SE}}(\hat{\theta})\right]$.

- **Ethics Case Clinic**

Short cases where AI output conflicts with community values. Students propose a **decision rule**, e.g., cost-sensitive loss

$$L = c_1 \mathbf{1}\{\text{FP}\} + c_2 \mathbf{1}\{\text{FN}\}$$

and justify c_1, c_2 for a chosen stakeholder.

- **Peer Review & Relational Intelligence**

Structured code/document reviews with **actionable feedback**; students track how revisions improved clarity and fairness.

6.3 Agility — Activity Ideas

- **Concept Drift Response Drill**

Provide train/test from P_0 ; then reveal test-2 from P_1 with drift. Students must: detect drift, re-tune, and justify a deployment gate (e.g., **stop** if AUC drops $> 5\%$).

- **Live Data Hackday**

Pull an open API; mid-day, change the KPI. Students pivot, update pipeline, and present a **retro** on trade-offs.

- **Rapid Experimentation Plan**

Students design an A/B test with power calculation

$$n = 2 \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{\delta/\sigma} \right)^2,$$

and discuss operational risks (ethics, data quality, monitoring).

7 Alignment with CSUEB Programs

- **Statistics B.S.** — Embed AI-assisted EDA, ethics mini-prompts, and agility sprints in regression, probability, and capstone.

- **Statistics M.S.** — Add concept-drift labs, model-risk statements, and stakeholder briefs in applied projects.
- **Data Science B.S.** — Use prompt-to-pipeline, reproducible MLOps practices, and live API hackdays across the curriculum.

8 Implementation Timeline (One Semester)

- **Weeks 1–2:** Publish AI policy & rubrics; run AI-fluency micro-labs.
- **Weeks 3–7:** Launch first team project with **stakeholder brief** + ethics prompts; add peer reviews.
- **Weeks 8–10:** Agility sprint with mid-course requirement change; model-change drill.
- **Weeks 11–14:** Capstone-style project; include **experience-gap** elements (deadlines, role rotation, external feedback).
- **Finals:** Oral validations; reflective memo on **AI use**, **verification**, and **professional growth**.

9 References

- Provost Muscat: *Back to the Bay Keynote Script – AI + Workforce Development (Aug 13–14, 2025)*. <https://www.csueastbay.edu/oaa/files/docs/provost-muscat-b2b-keynote.pdf>
- CSU East Bay — **Statistics B.S.:** https://catalog.csueastbay.edu/preview_program.php?catoid=39&poid=18177&hl=%22Statistics%22&returnto=search
- CSU East Bay — **Statistics M.S.:** https://catalog.csueastbay.edu/preview_program.php?catoid=39&poid=18257&hl=%22Statistics%22&returnto=search
- CSU East Bay — **Data Science B.S.:** https://catalog.csueastbay.edu/preview_program.php?catoid=39&poid=18358&hl=%22Data+Science%22&returnto=search